30 An electrical engineer has made a pair of components P and Q.

The resistance of P decreases as the temperature rises. The resistance of Q increases as the temperature rises.

The total resistance of P and Q in series remains constant when the temperature changes. The two components are connected in series with a power source supplying a constant current.

Which statement is correct?

A If a temperature change causes the potential difference (p.d.) across P to double, the p.d. across Q will always halve.

B The difference between the p.d. across P and the p.d. across Q does not change with temperature.

C The p.d. across component P increases as the temperature rises.

D The sum of the p.d. across P and the p.d. across Q does not change with temperature.